

Module 1: Web crawling

Module content

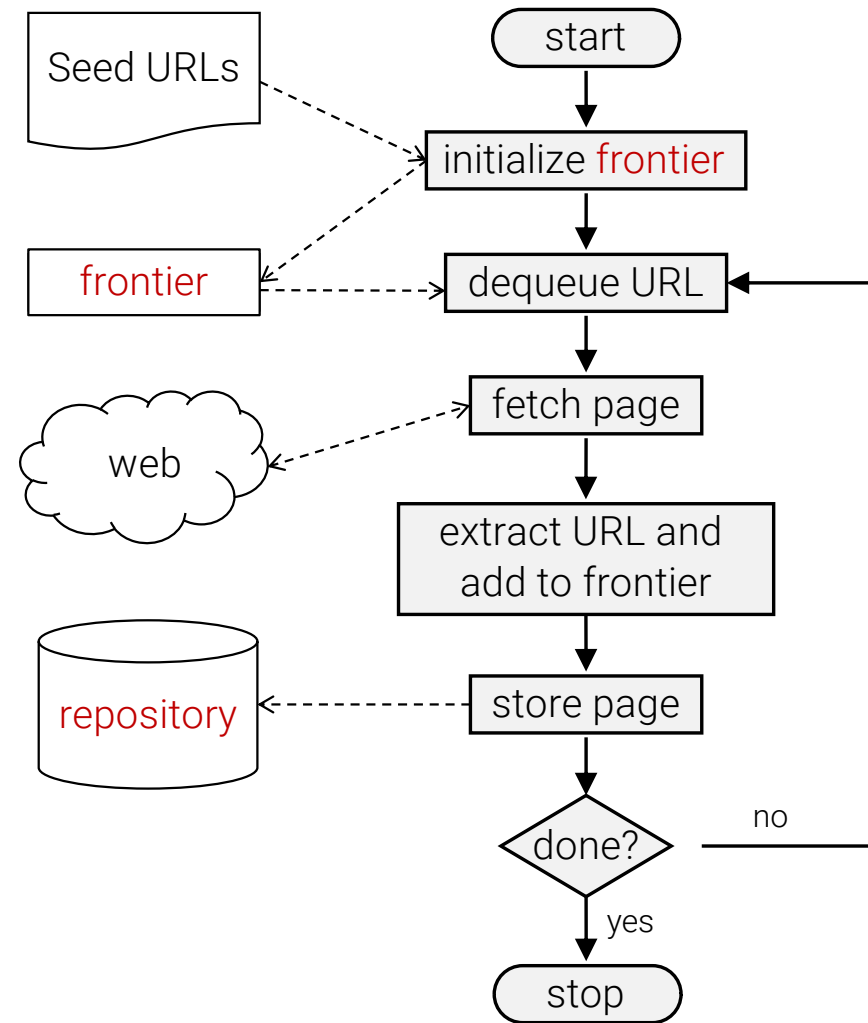
- Architecture
 - Basic
 - Parallel
 - Scalable
- Types
 - Universal
 - Focused
 - Topic-based
- Ethics

Web crawlers

- Description
- Challenges
- Application
- Abuse

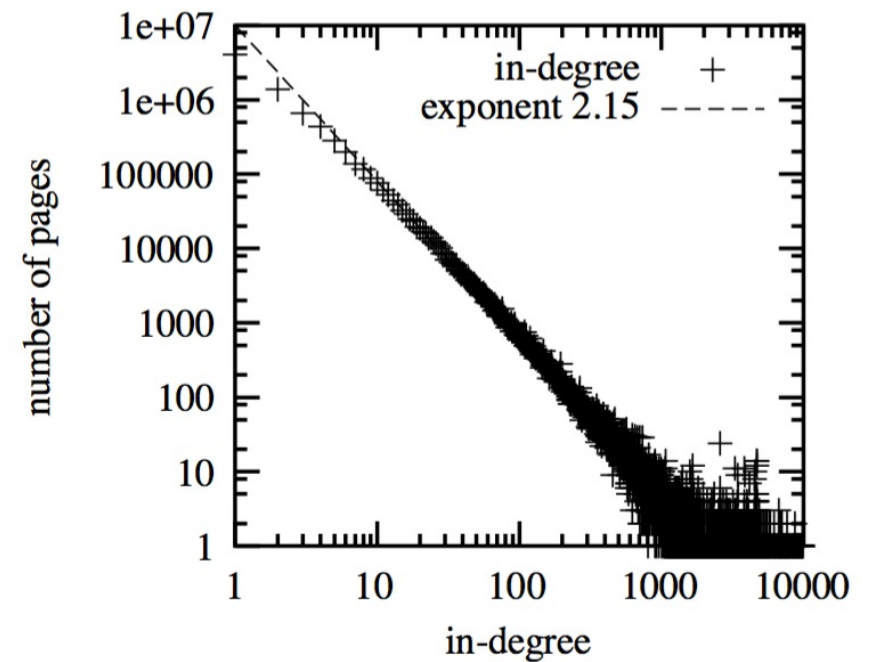
Basic architecture

- Frontier
- Seed pages
- Repository
- Implementation decisions:
 - Storage for URLs
 - URL selection strategy
 - Strategy if Frontier full
- Challenges and limitations



Breadth-first crawlers

- Frontier implementation (FIFO)
- Size limitations of frontier
- Is breadth-first crawling **random**?



Crawling history

- Reasons
- Implementation (disk, memory, hash tables)
- Duplicate detection (separate list)
- Time complexity

Preferential crawlers

- Preferential crawling
- Important
 - Preference setting (web topology, page content)
 - Selection of seed pages
- Implementation (sorted dynamic list + hash table)
- Time complexity
- Best-first crawling

Implementation details

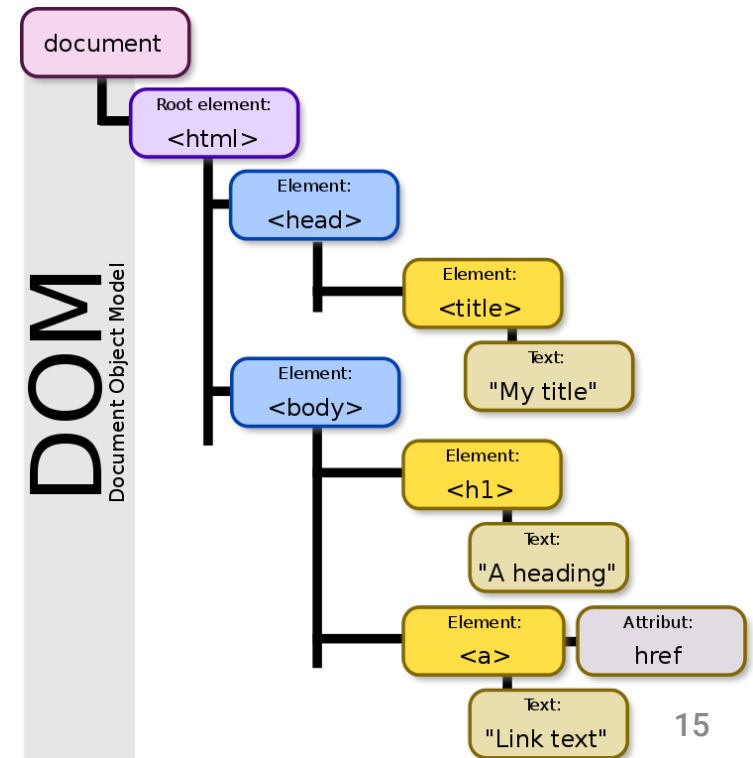
- Fetching
- Parsing
- Stopwords removal and stemming
- Link extraction and canonicalization
- Spider Traps
- Page repository
- Concurrency

Fetching

- Implementation (web client, HTTP requests)
- Limiting response time and page size
- HTTP headers (response code, redirect, last modified...)
- Error-checking and exception handling
- Keeping statistics
- Program language support

Parsing

- Parsing HTTP payload (link extraction, content analysis,...)
- Document Object Model (DOM)
- Cleaning HTML code prior parsing
- Basic format: HTML
- Other formats:
 - PDF, plain text, MS Power Point, MS Word ...
 - Flash, SVG, AJAX, XML-based ...



Stopword removal and stemming

- Stopwords (common words)
 - articles, conjunctions ...
- Stemming (removing inflated parts of words)
 - run, runs, running → run
 - argue, argues, arguing → argu
- Stemming ≠ lemmatization

Link extraction and canonicalization

- HTML link representation

`<a href="http://datalab.si/">Laboratory for Data Technologies</a>`

- Filtering

- Content type, HTTP head request
- Dynamic pages

- Autonomous querying (hidden web)

- URLs need to be absolute

- Canonicalization

Description and transformation	Example and canonical form
Default port number Remove	http://cs.indiana.edu:80/ http://cs.indiana.edu/
Root directory Add trailing slash	http://cs.indiana.edu http://cs.indiana.edu/
Guessed directory* Add trailing slash	http://cs.indiana.edu/People http://cs.indiana.edu/People/
Fragment Remove	http://cs.indiana.edu/faq.html#3 http://cs.indiana.edu/faq.html
Current or parent directory Resolve path	http://cs.indiana.edu/a/./../b/ http://cs.indiana.edu/b/
Default filename* Remove	http://cs.indiana.edu/index.html http://cs.indiana.edu/
Needlessly encoded characters Decode	http://cs.indiana.edu/%7Efil/ http://cs.indiana.edu/~fil/
Disallowed characters Encode	http://cs.indiana.edu/My File.htm http://cs.indiana.edu/My%20File.htm
Mixed/upper-case host names Lower-case	http://CS.INDIANA.EDU/People/ http://cs.indiana.edu/People/

Spider traps

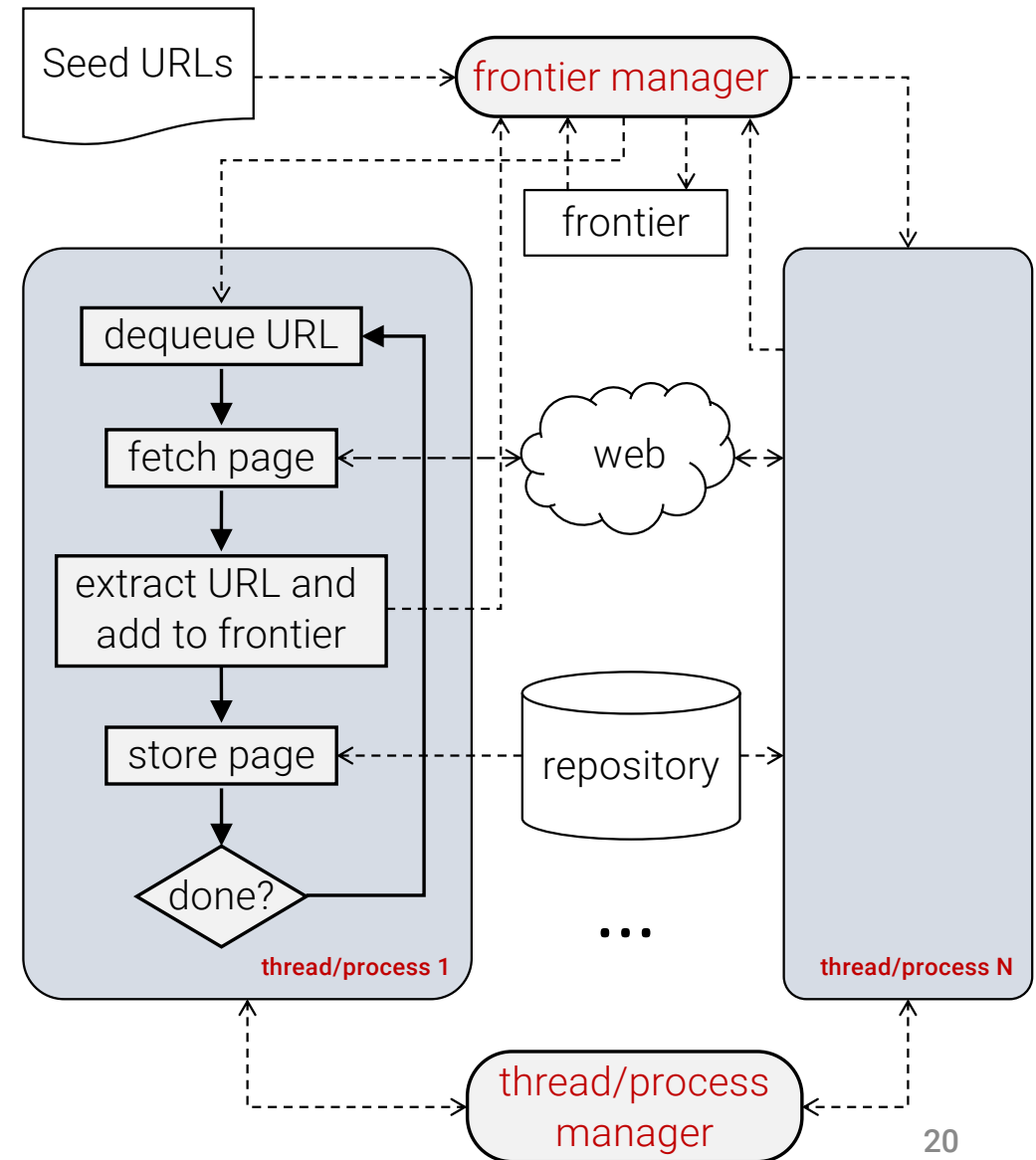
- Changing URLs of dynamically created links
- Same pages with new URLs
- Negative side effects
 - for crawlers (wasted bandwidth and disk space)
 - for servers (unintended DOS)
- Avoiding spider traps
 - refusing sending/receiving cookies,
 - limiting URL length
 - limiting number of pages from a given domain

Page repository

- File system
 - each page in a separate file
 - many pages in one file (plus hash table)
- Databases
 - embedded, e.g. Berkeley DB (key value store)

Concurrency

- Problematical resources
 - network,
 - CPU,
 - disk
- Improvements of sequential crawler
 - multithreading
 - Multiprocessing
- Dealing with empty Frontier



Universal crawlers

- Performance
 - Hundreds of thousands of pages fetched and indexed per second
 - Scalability
- Policy
 - Balancing coverage, importance, freshness

Coverage vs. freshness vs. importance...

- Indexing entire internet
- Importance measures (in-degree, PageRank, ...)
- Do nothing approach
- Fetching statistically representative sample of Web

$$\Pr(\textit{accept}(p) | \textit{crawl}(p)) * \Pr(\textit{crawl}(p))$$

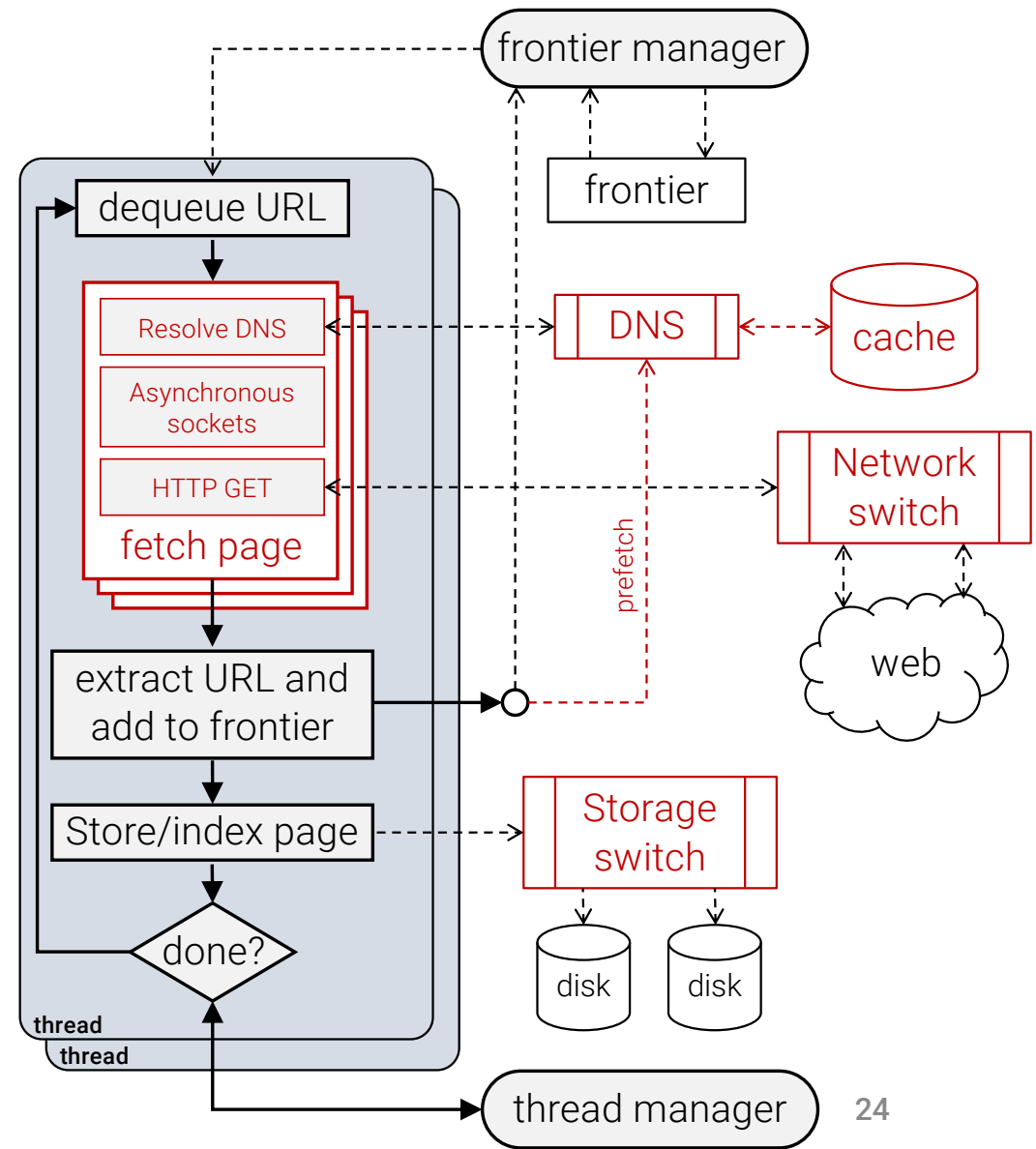
Coverage vs. freshness vs. importance*

- Coverage vs. freshness
- Dynamics of the Web (content, link structure)
 - 8% new pages every week
 - 62% of content is new
 - 25% new links per week
 - Once published, pages rarely change
 - Degree of change better predictor of future change than frequency of change

*Ntoulas, A., J. Cho, and C. Olston. *What's new on the web?: the evolution of the web from a search engine perspective*. In Proc. of International Conference on World Wide Web (WWW-2004), 2004.

Scalability

- Asynchronous sockets
- Multiple URL queues
- Working with DNS
 - Connections over UDP
 - employing fast in-memory cache
 - prefetching URLs
- Multiple network connections
- Use of SAN



Apache Nutch

- Main characteristics

razširljiv

lahko dela na porazdeljenih arhitekturah

napisan v javi

neodvisne podatkovne strukture

- Components

sestavljen iz petih komponent

pajek, iskalnik, indeksirnik, razčlenjevalnik, shramba

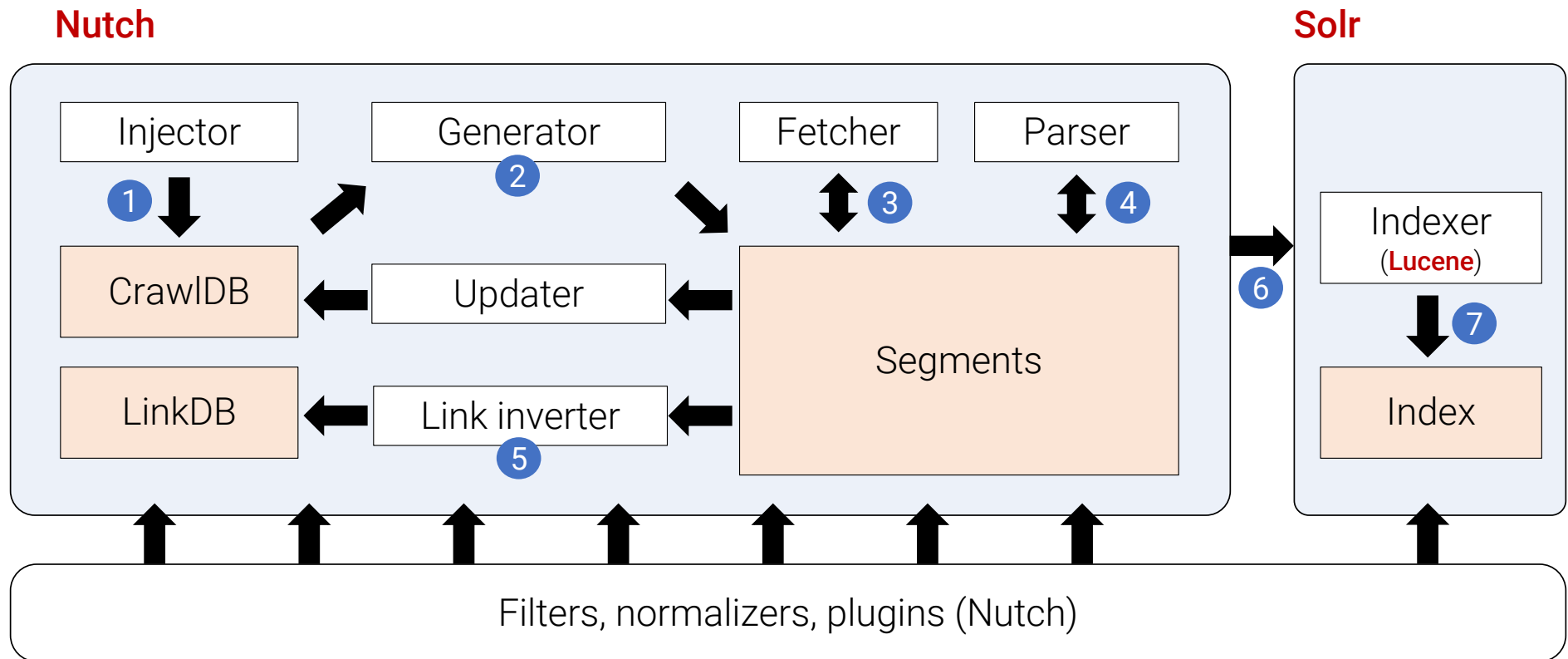
- Usage

- Branches

dve glavni veji: 1.x in 2.x (Apache Gora)

<http://nutch.apache.org/>

Apache Nutch architecture



Focused crawlers

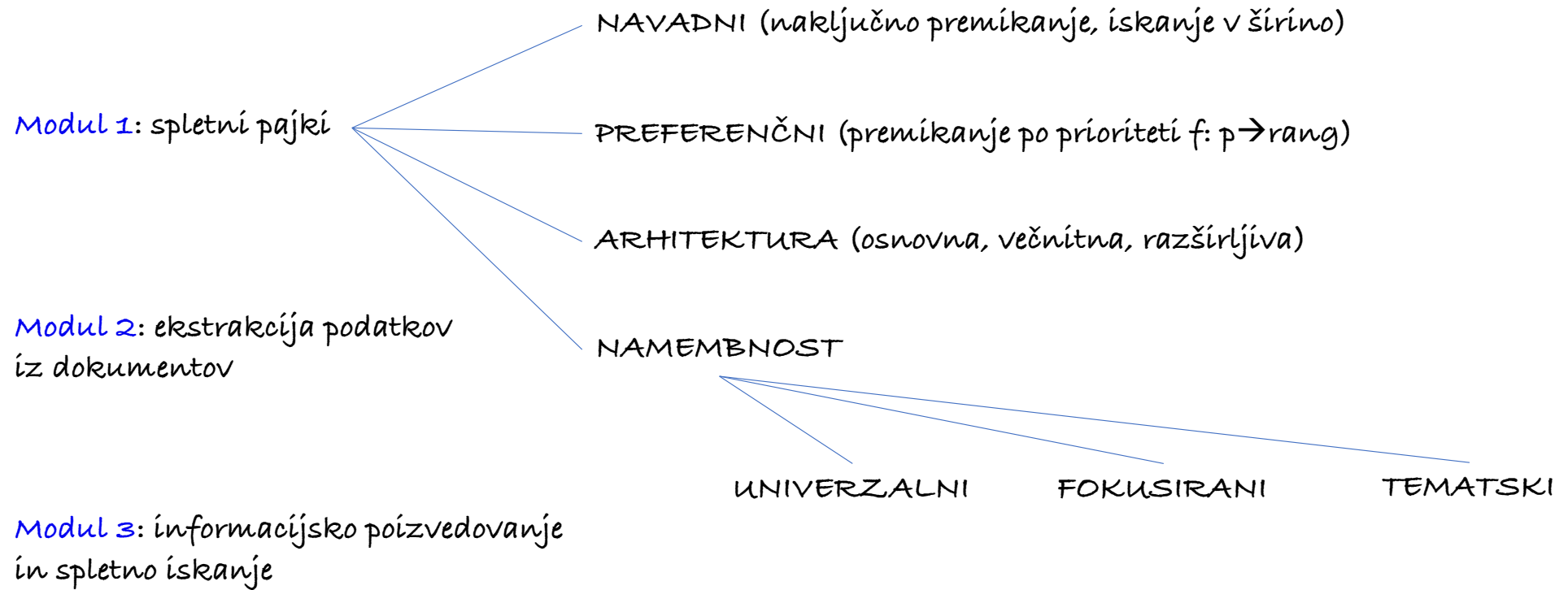
- Crawling pages of interest
- Taxonomy tree



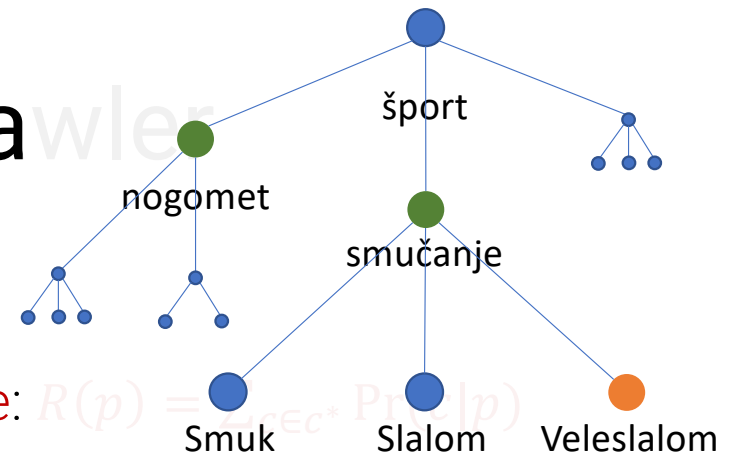
Web Directory of High-quality Resources

Arts Movies , Television , Music ...	Business Jobs , Real Estate , Investing ...	Computers Internet , Software , Hardware ...
Games Video Games , RPGs , Gambling ...	Health Fitness , Medicine , Alternative ...	Home Family , Consumers , Cooking ...
Kids and Teens Arts , School Time , Teen Life ...	News Media , Newspapers , Weather ...	Recreation Travel , Food , Outdoors , Humor ...
Reference Maps , Education , Libraries ...	Regional US , Canada , UK , Europe ...	Science Biology , Psychology , Physics ...
Shopping Clothing , Food , Gifts ...	Society People , Religion , Issues ...	Sports Baseball , Basketball , Soccer ...
World Deutsch , Français , 日本語 , Italiano , Español , Русский , Nederlands , Polski , Türkçe , Dansk , 简体中文 ...		

<http://odp.org/>



Classifier based focused crawler



■ Classification

- Each crawled page is assigned **relevance score**:
- where c^* denotes categories of interest

■ Soft focused strategy

- Links of page p are given preference $R(p)$ when written to the Frontier.

■ Hard focused strategy

- Find leaf category c_{leaf} that most likely include p
- If ancestor of c_{leaf} focus category $\rightarrow$ add links(p) to Frontier (using $R(p)$).

Text classification with naïve Bayes

$$\Pr(c|d) = \frac{\Pr(d|c) * \Pr(c)}{\Pr(d)}$$

where $c \in \mathcal{C}$ is a category from a set of categories $\mathcal{C}$ and d represents the crawled page. If d represented using multinomial document model then:

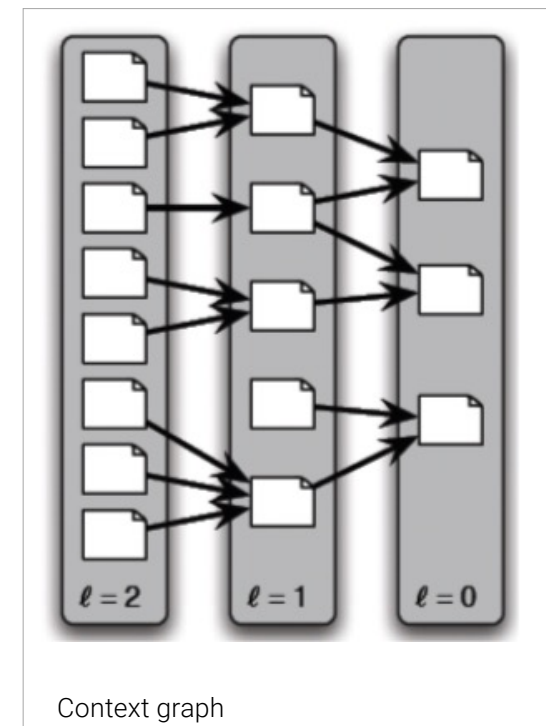
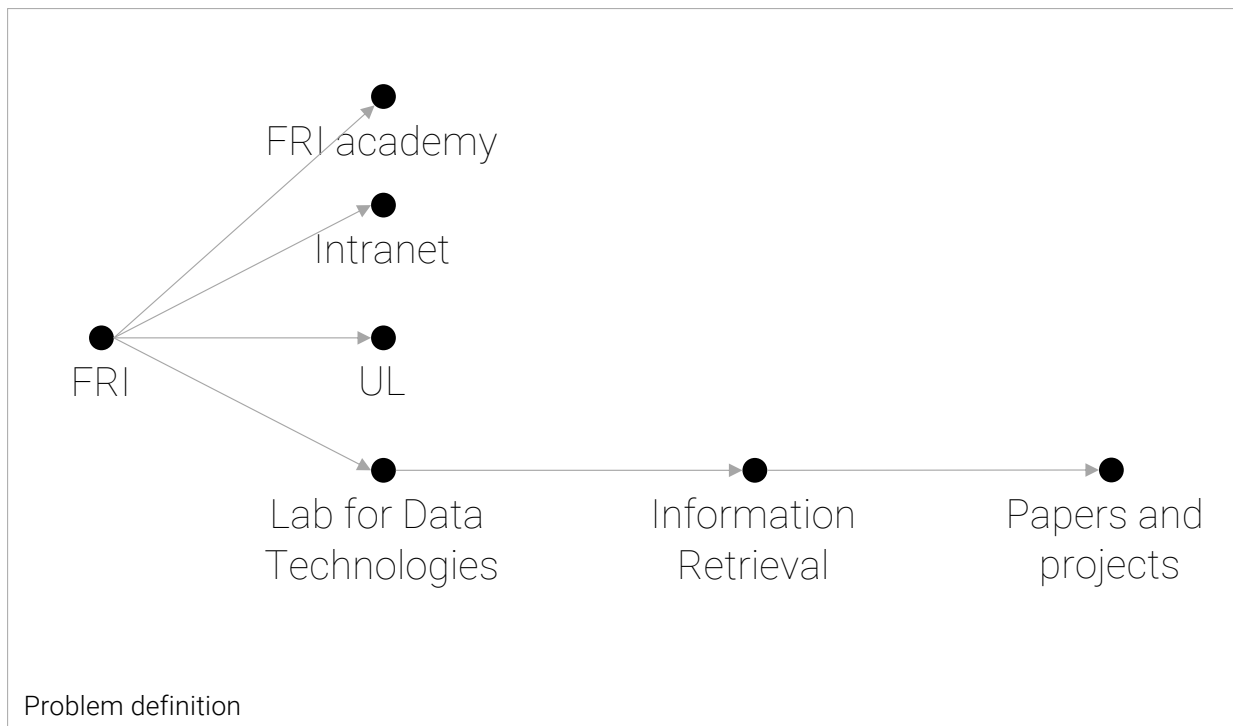
$\Pr(c|d) \approx \Pr(c) \prod_{i=1}^{|V|} \Pr(w_i|c)^{f_{w_i}}$, where V is the vocabulary of distinct words w_i in document d that we would like to classify and f_{w_i} is the frequency of words w_i in d . We are looking for $c_j \in \mathcal{C}$ where

$$c_j = \underset{c_k \in \mathcal{C}}{\operatorname{argmax}} \Pr(c_k) \prod_{i=1}^{|V|} \Pr(w_i|c_k)^{f_{w_i}}$$

Exercise

- Taxonomy: c_1 =sport, c_2 =politics
- Training dataset:
 - “A great game” → sports
 - “The election was over” → politics
 - “Very clean match” → sports
 - “A clean but forgettable game” → sports
 - “It was a close election” → politics
- Is the following text about sports or politics?
 - “A very close game” → ?

Context-based focused crawler ...



Context-based focused crawler*

■ Procedure:

- Use TFIDF to build vocabulary V of important words w for the topic of interest.
- Build Naïve Bayes classifiers for each layer: $\Pr(l_x|p) = \Pr(w_1|l_x) * \Pr(w_2|l_x) * \dots * \Pr(w_n|l_x) * \Pr(l_x)$, where $w_{1\dots n} \in (p \cap V)$
- If probability of layer with $\max \Pr(l|p) < threshold \rightarrow$ classify page p to *others* otherwise put to queue of the corresponding layer.
- Link to be followed next is selected from a nonempty queue of the lowest layer number.

*Diligenti, M., F. Coetzee, S. Lawrence, C. Giles, and M. Gori. *Focused crawling using context graphs*. In Proceedings of International Conference on Very Large Data Bases (VLDB-2000), 2000.

Tunneling...

- Limitations of Best-first approach
- Tunneling
 - Which links from irrelevant pages to follow?
 - How long to follow a tunnel?
 - How to estimate the potential of a certain tunnel?

Tunneling...

■ Concepts*:

- **Nuggets** and **duds**: relevant and irrelevant pages
- **Path**: a sequence from one nugget to another one
- **Threshold**: minimal similarity with topics to consider p as a nugget
- **Cutoff**: max length of a path
- **Distance**: length from the last nugget

$$distance = \begin{cases} 0, & \text{if } p \text{ nugget} \\ 1 + distance(parent(p)), & \text{otherwise} \end{cases}$$

*Bergmark, D., Lagoze, C., & Sbityakov, A. (2002), *Focused Crawls, Tunneling, and Digital Libraries*, In proc. of the European Conference on Digital Libraries (ECDL) 2002, Rome, September 2002, 91-106.

Tunneling

- Experimental findings (on 500.000 documents)
 - Most common distances are 7 and 8
 - Better parents have better children, but the opposite is not true.
 - For high-scoring parents, highly scoring grandparents important.

Distance incorporating path history

$$distance = \begin{cases} 0, & \text{if } p \text{ nugget} \\ \min(1, (1 - c)e^{\frac{2d_p}{C}}), & \text{otherwise} \end{cases}$$

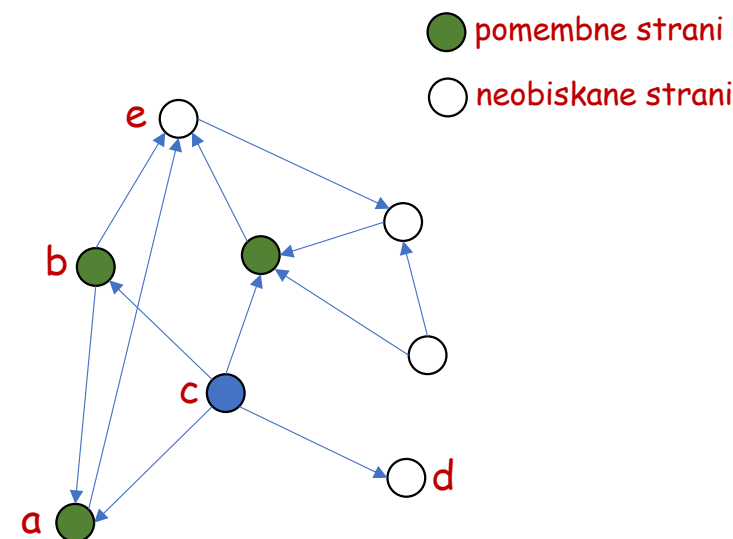
where d_p : parent distance, c : current node correlation, C : cutoff

Topical crawlers

- Limited prior knowledge → real time crawling (no index), calculating relevance on-the-fly
- Pros:
 - fresh hits (access to documents not yet indexed)
- Cons:
 - slower than index-based search,
 - less accurate ranking (no global measures)
- Cues to explore and other influencing parameters

Cues from words and hyperlinks*

- Lexical metrics
- Link metrics:
 - distance*
 - co-citation (semantic similarity)
 - hubs
 - bibliographic coupling



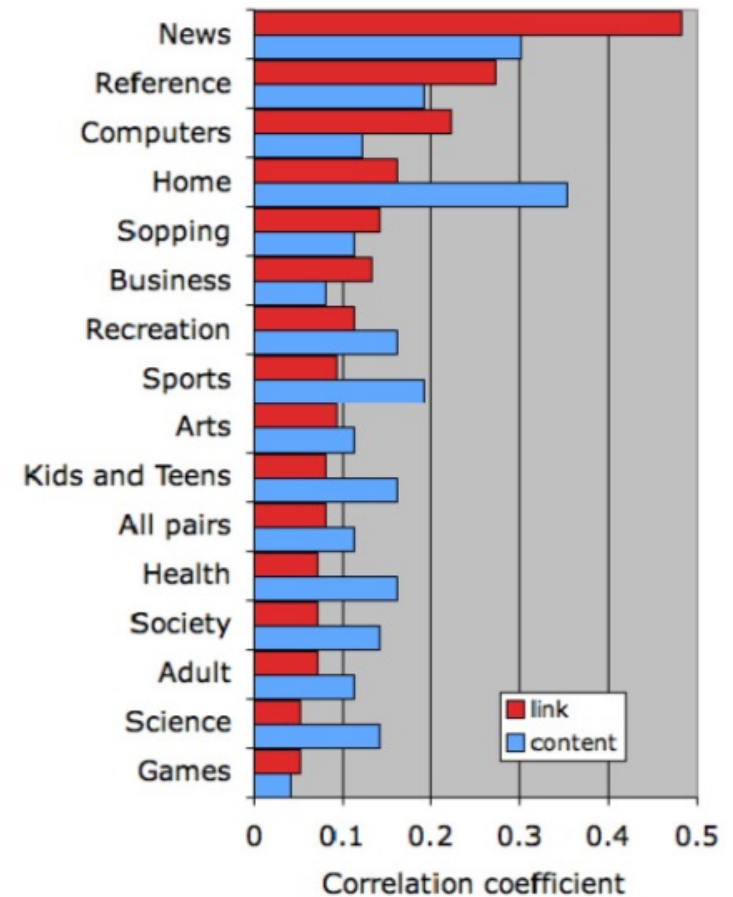
ko-citiranja, stičišča, bibliografska povezanost

*Menczer, F. *Lexical and semantic clustering by web links*
Journal of the American Society for Information Science
and Technology, 2004, 55(14): p. 1261-1269.

Link content and cluster conjecture*

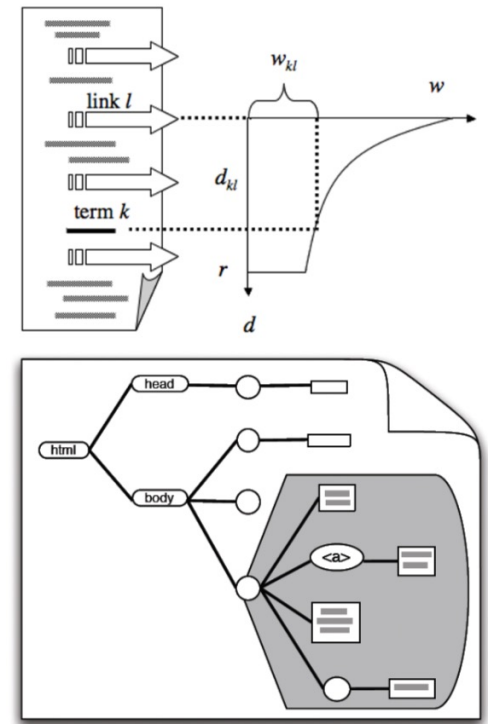
- Correlation between similarity measures:
 - content-based,
 - link-based (bag of links), and
 - semantic-based (ODP – ground truth).

*Menczer, F. *Mapping the semantics of web text and links*.
Internet Computing, IEEE, 2005, 9(3): p. 27-36.



Variations of best-first crawlers

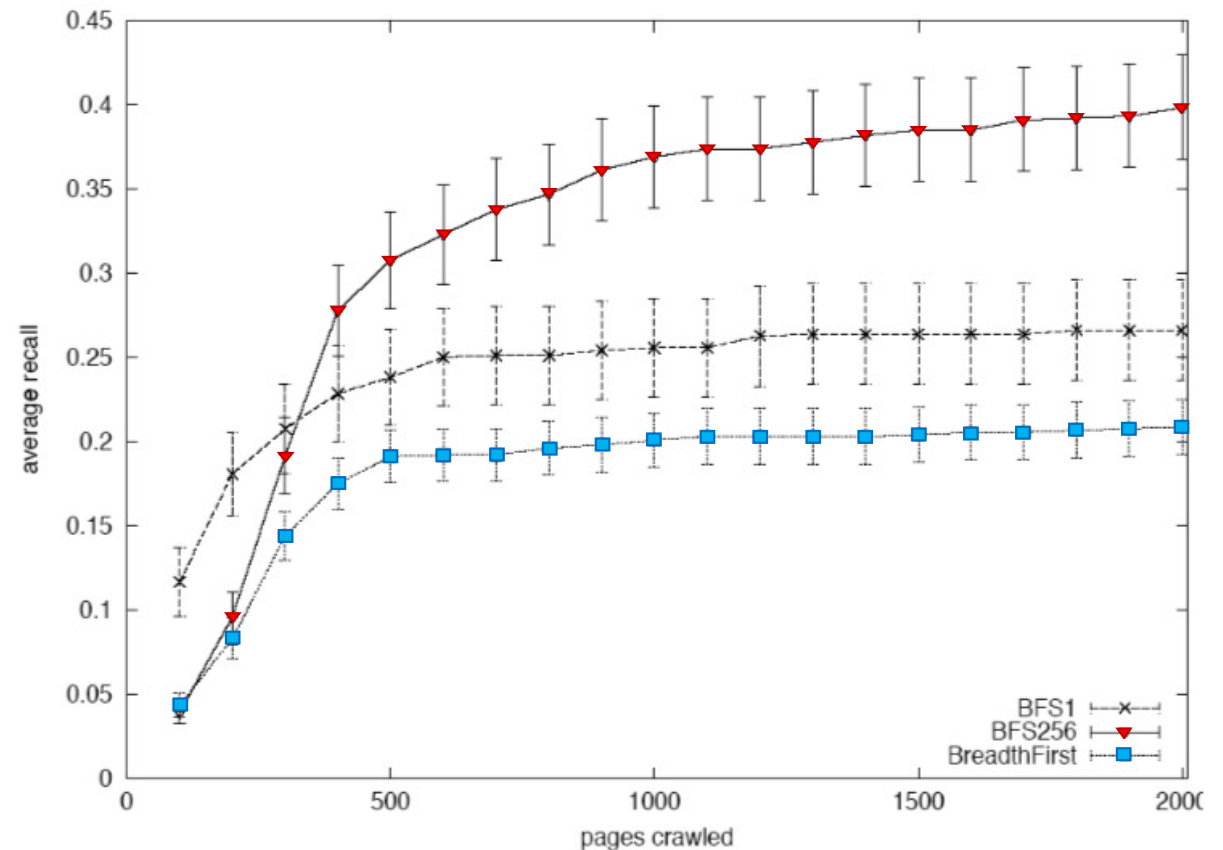
- Naïve best-first crawlers
- InfoSpiders [1]
- Clever crawler [2]
- SharkSearch [3]



1. Menczer, F. and R. Belew. *Adaptive retrieval agents: Internalizing local context and scaling up to the Web*. **Machine Learning**, 2000, 39(2): p. 203- 242..
2. Chakrabarti, S., B. Dom, S. Kumar, P. Raghavan, S. Rajagopalan, A. Tomkins, D. Gibson, and J. Kleinberg. *Mining the Web's link structure*. **Computer**, 2002, 32(8): p. 60-67.
3. Hersovici, M., M. Jacovi, Y. Maarek, D. Pelleg, M. Shtalhaim, and S. Ur. *The shark-search algorithm. An application: tailored Web site mapping*. **Computer Networks**, 1998, 30(1-7): p. 317-326.

Best-N-first crawler

- Visiting of less promising pages to avoid “local optimums”
- Best-N-first crawler
 - follow best N leads
 - fetch all, merge-sort, repeat
 - Exploitation vs exploration



Adaptive topical crawlers

- Static vs adaptive strategy (incremental learning)
- Example of adaptive strategy:

```
select features  $fe_i \in F$ 
while not enough relevant pages crawled do
  crawl new page  $p$ 
  calculate page priority  $P(p) = f(\lambda(fe_1), \lambda(fe_2), \dots, \lambda(fe_m))$ 
  on every  $n$  crawled pages
    recalculate interest factors  $\lambda(fe_i)$ 
    reorder frontier
end
```

Interest factor

$$\lambda(fe_m)$$

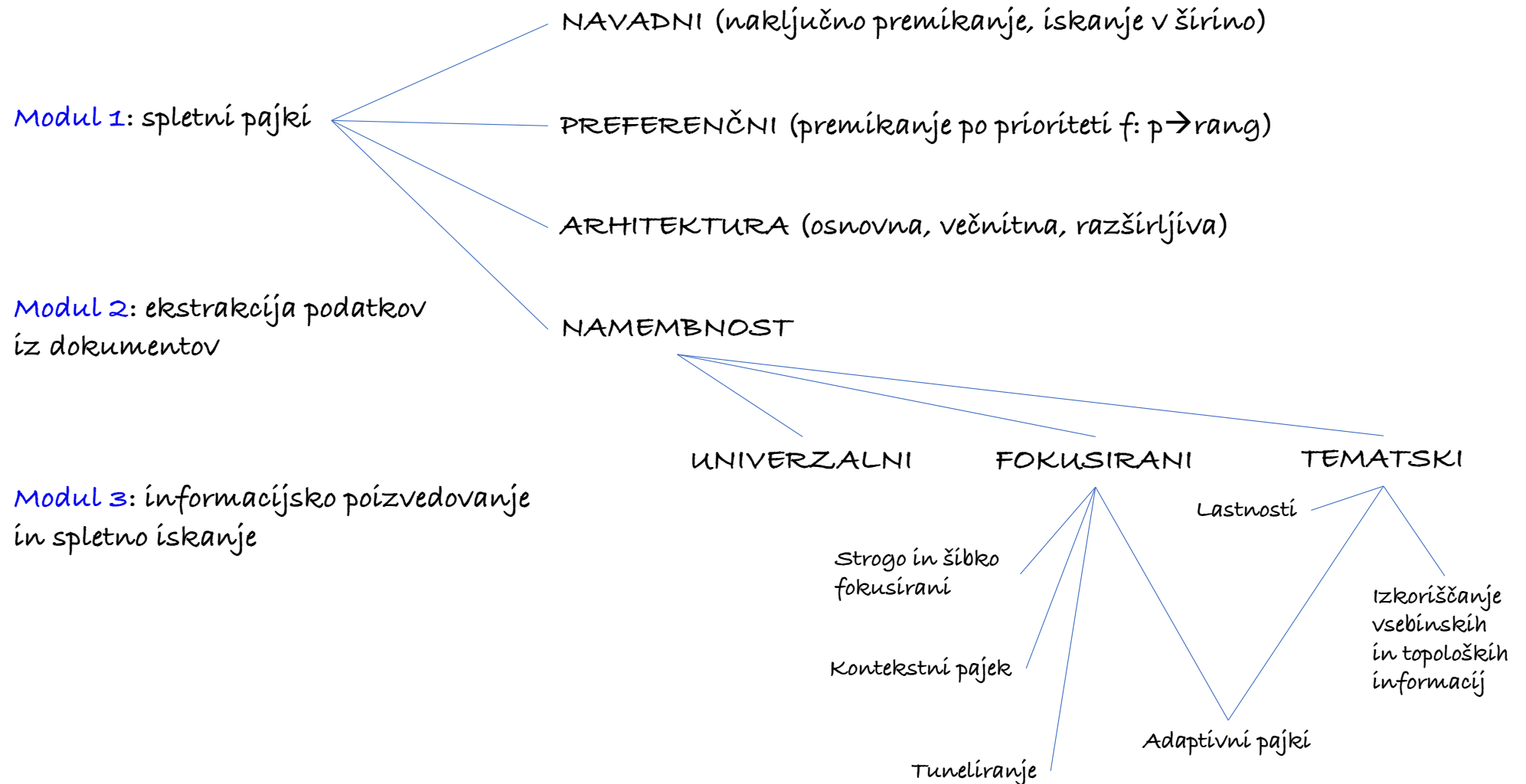
Intelligent crawler*

■ Features

- Content based $\rightarrow I_c(c)$
- URL token based $\rightarrow I_u(c)$
- Link based $\rightarrow I_i(c)$
- Sibling based $\rightarrow I_s(c)$

$$PriorityValue = w_c \log(I_c(c)) + w_u \log(I_u(c)) + w_i \log(I_i(c)) + w_s \log(I_s(c))$$

*Aggarwal, C., F. Al-Garawi, and P. Yu. *Intelligent crawling on the World Wide Web with arbitrary predicates*. In Proceedings of 10th International Conference on World Wide Web (WWW-2001), 2001.



Reinforcement learning (RL)

- Reinforcement learning (RL) is technique to find optimal solution by trial and error.
- RL imitates reward-motivated behavior.
- Simulation of real-life learning
 - try different strategies
 - learn from the feedback
 - act according to the best strategy



RL video demonstration



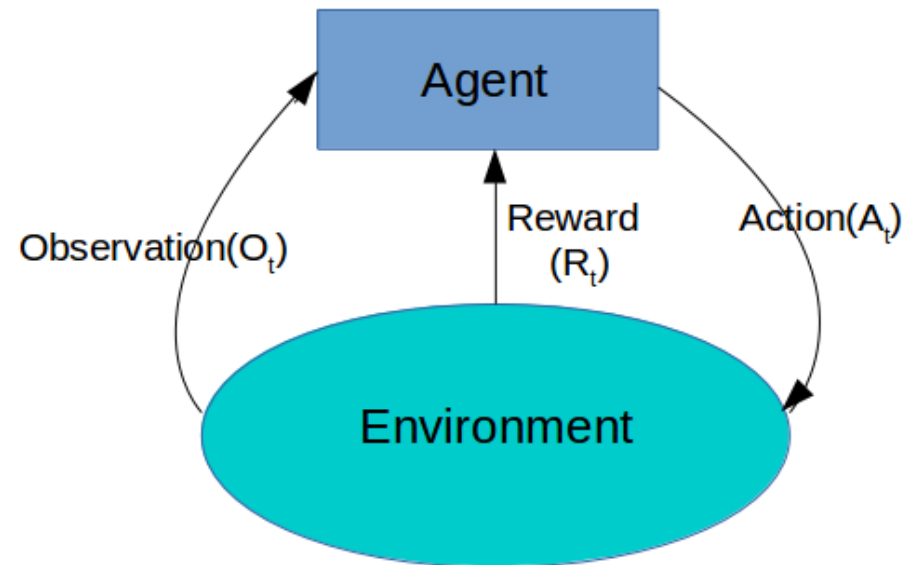
DRL

Deep Reinforcement Learning

Formalization

■ Key concepts in RL:

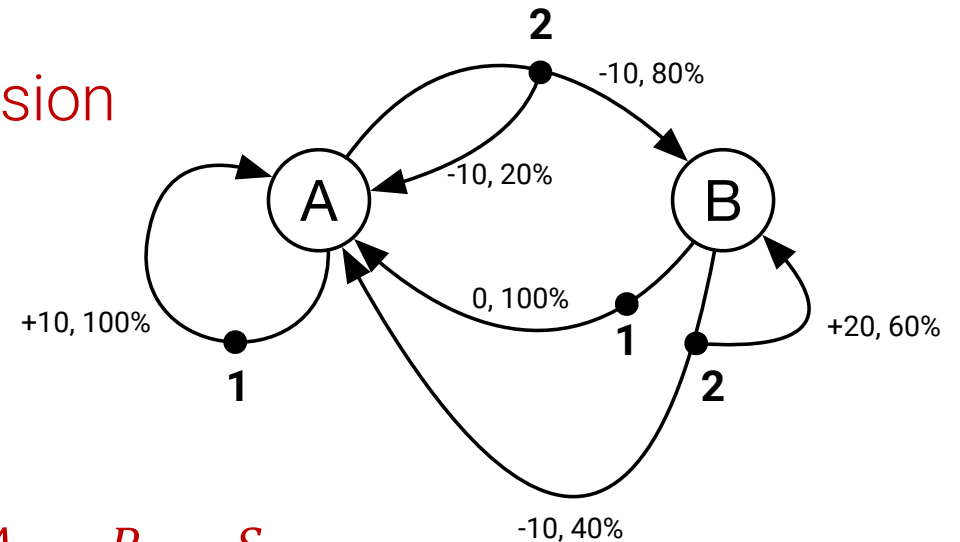
- agent,
- state,
- action,
- environment,
- reward.



RL – modeling environment

■ Environment: A finite Markov Decision Process (MDP)

- Discrete time $t = 1, 2, 3$
- A finite set of states
- A finite set of actions
- A finite set of rewards
- Life is trajectory $\dots S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1}, R_{t+2}, S_{t+2}, \dots$ with Markov dynamics:



$$p(r, s' | s, a) = \text{Prob}[R_{t+1} = r, S_{t+1} = s' | S_t = s, A_t = a]$$

RL – policies

- Policy π – defines for each state s what action a to take
- Deterministic policy: $a = \pi(s)$
- An agent following a policy: $A_t = \pi(S_t)$
- The agent's goal is to choose actions A_t in a way to maximize the **discounted sum** of future rewards:

$$R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

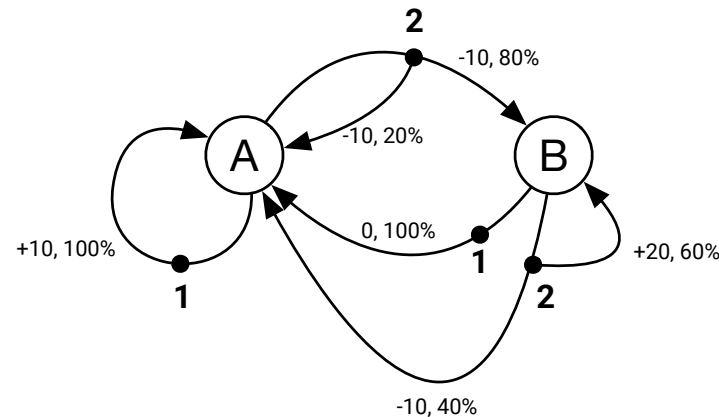
RL – Action-value functions

- An **action-value function** tells how good is to be in a state, take an action and then follow a policy:

$$q_{\pi}(s, a) \approx \mathbb{E}_{\pi}[G_t | S_t = s, A_t = a] = \mathbb{E}_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} | S_t = s, A_t = a \right]$$

Action-value function

State	Action	Value
A	1	130,39
A	2	133,77
B	1	166,23
B	2	146,23



RL – State value function

- A **state value function** tells, how good it is for the agent to be in a certain state.
- The value of a state s under a policy π , denoted $v_\pi(s)$, is the expected return when starting in s and following π thereafter.
- For MDPs, we can define v_π as:

$$v_\pi(s) \approx \mathbb{E}_\pi[G_t | S_t = s] = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} | S_t = s \right]$$

RL – Optimal policies...

- An optimal policy π_* is **better** than other policies. It maximizes the state and action-value functions:

$$q_{\pi_*}(s, a) = \max_{\pi} q_{\pi}(s, a), \quad v_{\pi_*}(s) = \max_{\pi} v_{\pi}(s)$$

- Bellman optimality equations:

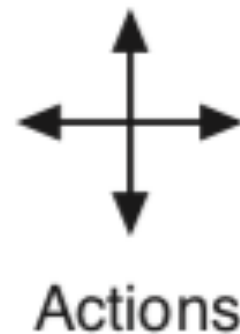
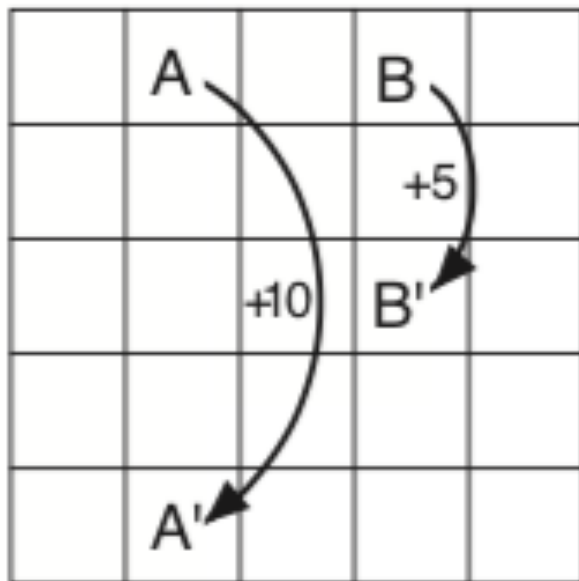
$$v_{\pi_*}(s) = \max_a q_{\pi_*}(s, a) = \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma v_{\pi_*}(s')],$$

$$q_{\pi_*}(s, a) = \sum_{s', r} p(s', r | s, a) \left[r + \gamma \max_{a'} q_{\pi_*}(s', a') \right]$$

RL – Optimal policies

- Given the optimal value functions, it is easy to act optimally:
 - If we have v_{π^*} then we just need to check which states that are reachable from the current state gives the best value.
 - If we have q_{π^*} then we just select the best action.
- Optimal policy is **greedy** with respect to the optimal value functions.

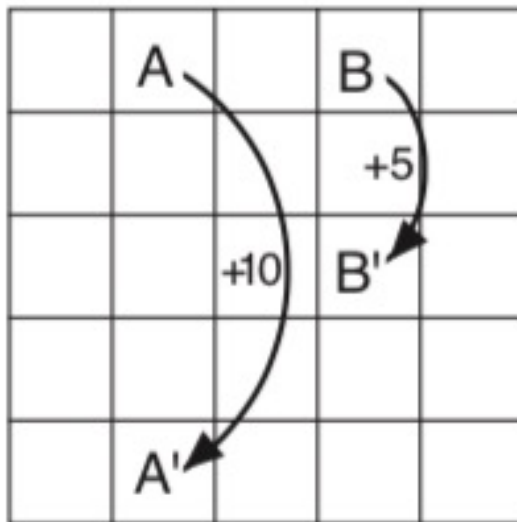
Gridworld example...



3.3	8.8	4.4	5.3	1.5
1.5	3.0	2.3	1.9	0.5
0.1	0.7	0.7	0.4	-0.4
-1.0	-0.4	-0.4	-0.6	-1.2
-1.9	-1.3	-1.2	-1.4	-2.0

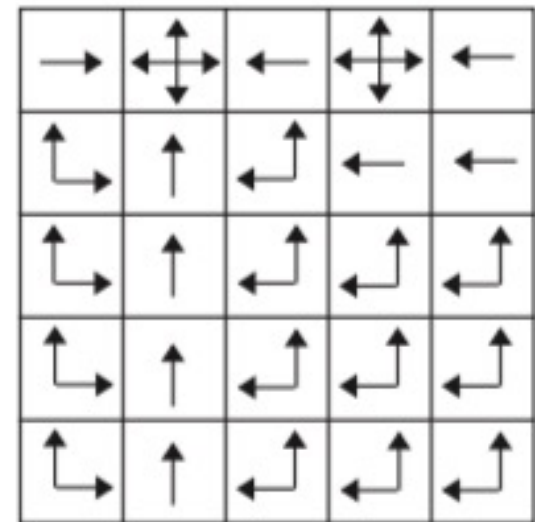
Random action policy

Gridworld example



22.0	24.4	22.0	19.4	17.5
19.8	22.0	19.8	17.8	16.0
17.8	19.8	17.8	16.0	14.4
16.0	17.8	16.0	14.4	13.0
14.4	16.0	14.4	13.0	11.7

Optimal state-value function



Optimal policy

Q-learning

- The simplest RL algorithm:

- Initialize an array $Q(s, a)$
- Choose actions in any way (e.g. to take all states several times)
- On each time step, change one element of the array:

$$\Delta Q(S_t, A_t) = \alpha \left(R_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t) \right)$$

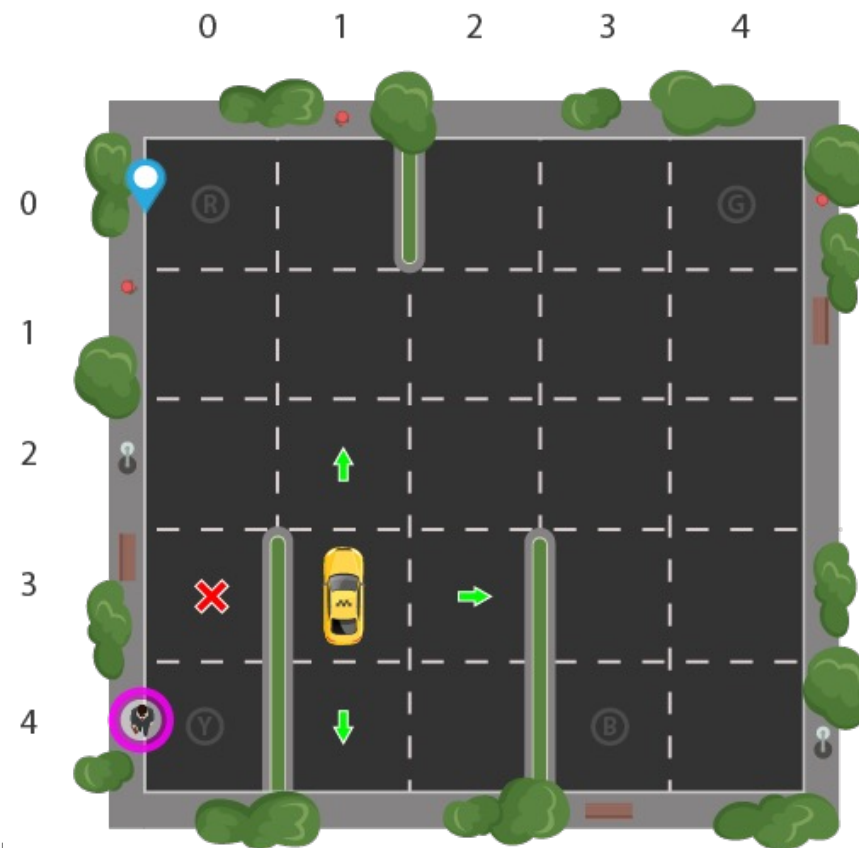
- If desired, reduce step-size parameter α over time.

- Theorem*: for appropriate choice of α , Q converges to q_{π^*} and its greedy policy to an optimal policy π_* (Watkins & Dayan, 1992).

*Watkins, C. J. C. H., Dayan, P. (1992). *Q-learning*. Machine Learning, 8:279–292.

Example of using Q-learning

- SmartCab



Reinforcement learning vs ML

- Specifics:
 - no supervisor,
 - importance of time,
 - delayed rewards,
 - agent action effects its next input.
- History abstraction (Markov state)
- Algorithms (Q-learning, SARSA, DQN, DDPG)

RL-based Focused Crawler*

- Goal of focused crawler
 - Collect relevant pages while avoid irrelevant (due to resources)
 - Selection of most promissing link may not lead to relevant pages in a long run.
- RL agent evaluates if a link selection can yield a long-term optimal reward...

*Han, M., Willemin, P., Senellart, P. *Focused Crawling Through Reinforcement Learning*, Lecture Notes in Computer Science, Springer International Publishing AG, T. Mikkonen et al. (Eds.): ICWE 2018, LNCS 10845, pp. 261–278, 2018.

Problem modelling

- Representation
 - States – web pages
 - Actions – hyperlinks

- Search space simplification:
 - Pages with same feature values represented as a single page
 - Links with same feature values represented as one link.

Features of pages and links

■ Pages

- Relevance to target topic and selected categories
- Relevance change
- Average relevance of parent pages
- Distance from the last relevant page

■ Links

- Relevance of target topic and selected categories,
- Average relevance of parents

MDPs with Prioritizing updates

- Using prioritized URLs, focused crawler would select best link to go – no matter of its current position!
- Crawling agent has to follow this principle.

- Prioritized updates

to update state s with change δ in $U(s)$:

update $U(s)$

priority of $s \leftarrow 0$

for each predecessor s' of s :

priority $s' \leftarrow \max$ of current and $\max_a \delta \hat{T}(s', as')$

Linear Function Approximation

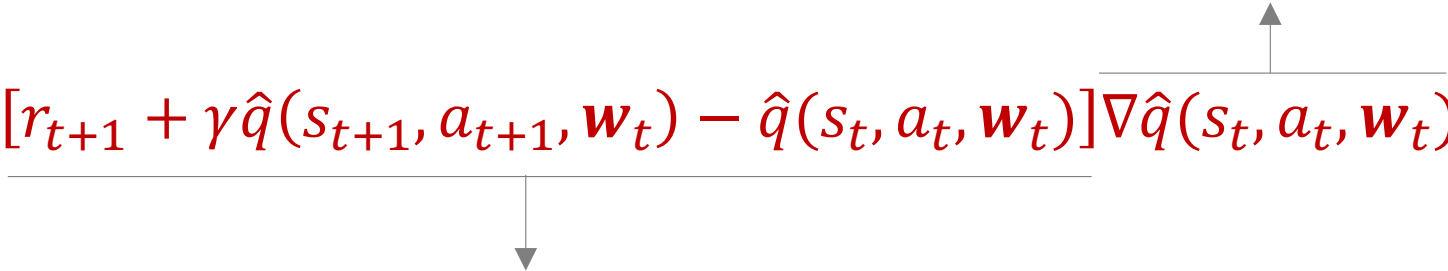
- In **large state-action spaces**, a tabular representation of action-value functions impossible.
- Action values are represented with **linear function approximation**:

$$\hat{q}(s, a, \mathbf{w}) = \mathbf{w}^T \mathbf{x}(s, a) = \sum_{i=1}^d w_i x_i(s, a)$$

where $\mathbf{x}(s, a)$ is a feature vector and $\mathbf{w}$ weight vector.

Gradient descent update on SARSA

- At each time step, weight vector $\mathbf{w}$ is updated.
- Gradient descent method is used (in direction that minimizes TD error):

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha [r_{t+1} + \gamma \hat{q}(s_{t+1}, a_{t+1}, \mathbf{w}_t) - \hat{q}(s_t, a_t, \mathbf{w}_t)] \nabla \hat{q}(s_t, a_t, \mathbf{w}_t)$$


Partial derivatives of the function

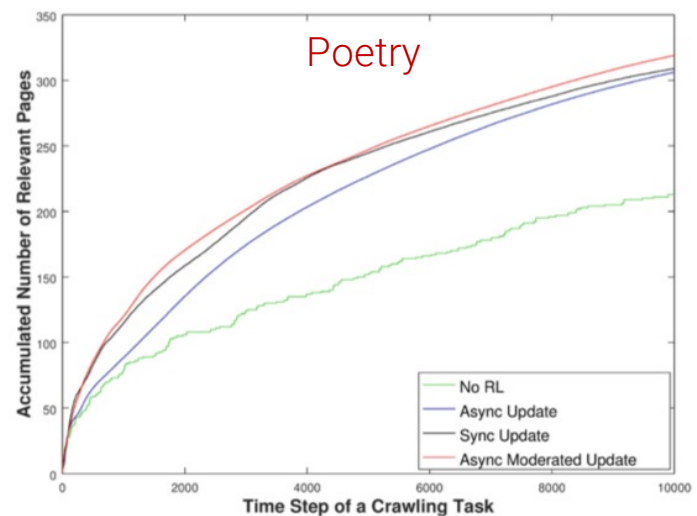
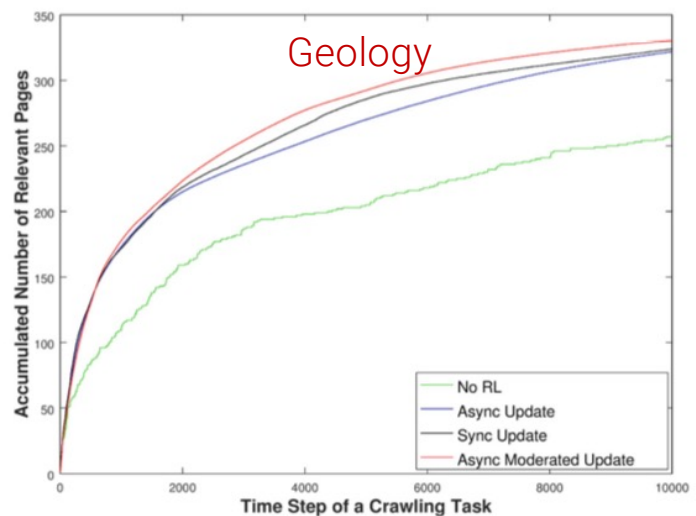
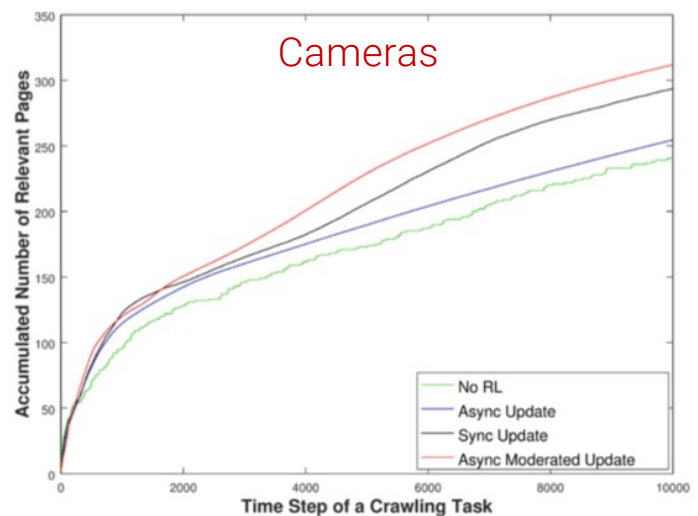
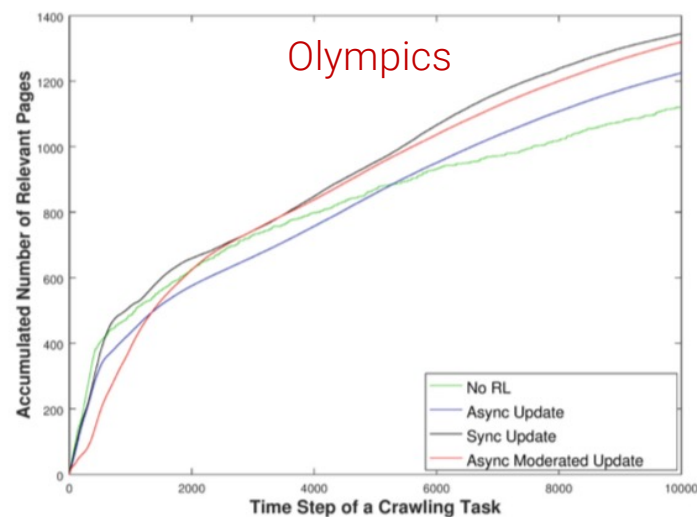
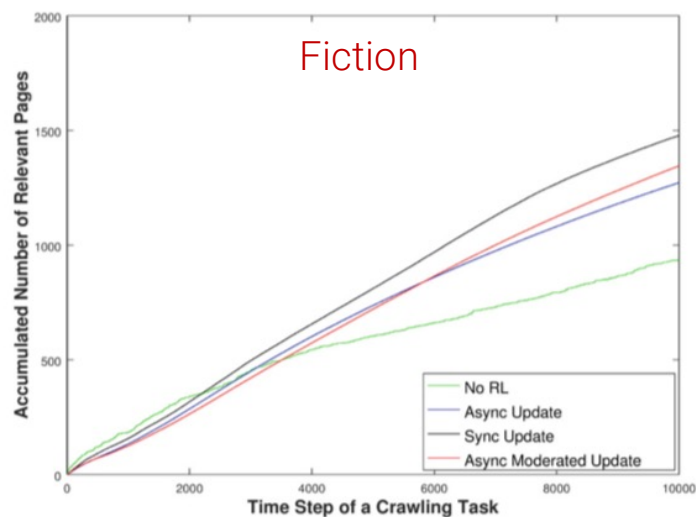
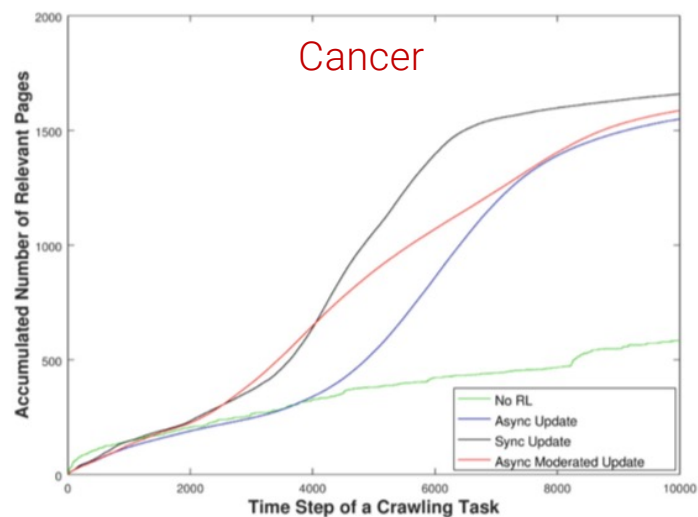
TD or Bellman error: difference between the action value in two successive states, t and $t+1$. "

Evaluation setting...

- Crawling task starts with a seed page and terminates when n pages are visited (n set to 10.000).
- At each time step, page is crawled and analyzed (cosine and word2vec similarity measures).
- Rewards:
 - 30 if TFIDS similarity > 0
 - 30 if TFIDF=0 and word2vec $> 0,5$
 - 20 if TFIDF=0 and word2vec $> 0,4$
 - -1, otherwise

Evaluation setting

- **Environment**: database dump of Simple English Wikipedia.
- **Target topics**:
 - Representative: Fiction, Olympics, Cancer
 - Sparse: Cameras, Geology, Poetry
- **Seed pages**: main pages of each category
- **Parameters**:
 - $\varepsilon = 0.1$, discount rate $\gamma = 0.9$, step size $\alpha = 0.001$.



Accumulated number of relevant pages per time step

Evaluation of crawlers

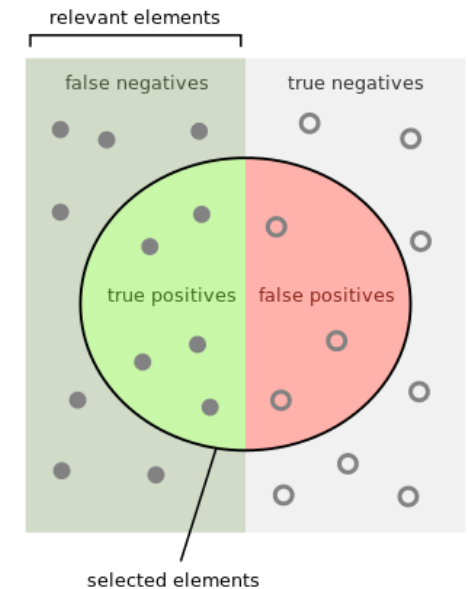
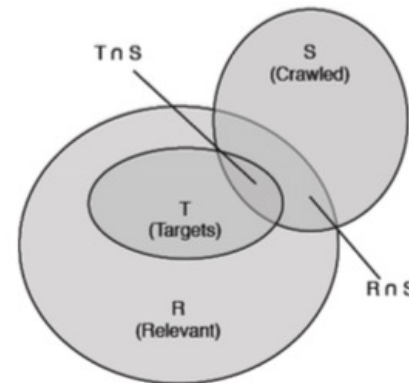
- How to compare crawling algorithms?
 - comparing applications
 - comparing algorithms
- Requirements:
 - sufficient number of runs over different topics
 - sound methodologies to consider temporal nature of results
- Challenges
 - unavailability of relevant sets for particular topics/queries.

Relevance estimation

- Estimation of page relevance (and quality)
 - number of keywords in the page
 - frequency of keywords
 - cosine similarity
 - classifier-based
 - N crawlers from same seeds until p pages retrieved
 - PageRank, HITS

Metrics

- Precision-like metrics:
 - harvest rate for Boolean relevance
 - average relevance for continuous relevance scores
- Search length
- Recall-like metrics:
 - without normalization
 - on predesigned crawling experiment (seeds selection)
- Use of critical resources



How many selected items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

Crawler ethics

- Limiting server requests
- Distribution of request
- Crawler etiquette
- **User-Agent** HTTP and **From** header
- Robot Exclusion Protocol, **robots.txt**
 - User-agent: *
 - Disallow: /
- HTML meta tag

